

Отчет по лабораторной работе №3

Модификации градиентного спуска

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1 Постановка задачи

Цель работы — реализовать и исследовать модификации градиентного спуска, в которых используются инерция, накопление статистики градиентов и адаптивный масштаб шага по координатам.

В работе реализованы:

1. Momentum;
2. Nesterov accelerated gradient;
3. AdaGrad;
4. RMSProp;
5. AdaDelta;
6. Adam.

Во всех экспериментах использовалась фиксированная точность

$$\varepsilon = 10^{-8},$$

а критерием остановки была норма градиента:

$$\|\nabla f(x_k)\| \leq \varepsilon.$$

Промежуточный вывод. Третья лабораторная продолжает вторую: вместо чистого шага по антиградиенту используются правила, которые помнят прошлые шаги или прошлые градиенты. Поэтому здесь особенно важны гиперпараметры. Они задают, насколько сильно метод опирается на текущий градиент, накопленную скорость и координатные масштабы.

2 Используемые модели задач

Модели задач переиспользуются из лабораторной работы №2:

- хорошо обусловленная квадратичная функция

$$f(x, y) = \frac{1}{2}(x^2 + y^2);$$

- плохо обусловленная квадратичная функция

$$g(x, y) = \frac{1}{2}x^2 + 50y^2;$$

- функция Розенброка

$$r(x, y) = (1 - x)^2 + 100(y - x^2)^2;$$

- функция Экли;
- функция Химмельблау.

Для квадратичных функций использовалась стартовая точка $(-2, -2)$. Для сложных функций в конфигурации задавались несколько стартовых точек, чтобы проверить чувствительность методов к области притяжения.

Промежуточный вывод. Повторное использование моделей из второй лабораторной делает сравнение аккуратнее: меняется не задача, а только правило построения следующего приближения. Momentum, RMSProp и Adam берут тот же градиент, но по-разному превращают его в обновление с учетом истории. На одних и тех же поверхностях видно, где память метода действительно помогает, а где главная сложность остается в геометрии функции.

3 Описание реализованных методов

3.1 Momentum

Метод Momentum добавляет инерционный член:

$$v_t = \beta v_{t-1} + \alpha \nabla f(x_t),$$

$$x_{t+1} = x_t - v_t.$$

Параметр β отвечает за сохранение предыдущего направления движения.

Промежуточный вывод. Инерция ускоряет движение вдоль устойчивого направления: новый шаг складывается не только из текущего градиента, но и из накопленной скорости. Если несколько градиентов подряд смотрят примерно в одну сторону, скорость растет и метод быстрее проходит пологие

участки. Но около минимума или в узкой долине это может мешать: накопленная скорость продолжает тянуть точку вперед, хотя текущий градиент уже требует торможения или поворота.

3.2 Метод Нестерова

В методе Нестерова градиент вычисляется в точке предпросмотра:

$$\begin{aligned}\tilde{x}_t &= x_t - \beta v_{t-1}, \\ v_t &= \beta v_{t-1} + \alpha \nabla f(\tilde{x}_t), \\ x_{t+1} &= x_t - v_t.\end{aligned}$$

Промежуточный вывод. Метод Нестерова пытается заранее оценить, куда приведет инерция, поэтому градиент вычисляется не в текущей точке, а в точке предпросмотра. Такая структура делает корректировку направления более ранней: метод видит последствия накопленной скорости до фактического обновления. За это он платит дополнительным вычислением градиента на каждой итерации, что видно в счетчиках вызовов.

3.3 AdaGrad

AdaGrad накапливает сумму квадратов градиентов:

$$\begin{aligned}s_t &= s_{t-1} + \nabla f(x_t)^2, \\ x_{t+1} &= x_t - \alpha \frac{\nabla f(x_t)}{\sqrt{s_t} + \varepsilon_{\text{num}}}.\end{aligned}$$

Промежуточный вывод. AdaGrad уменьшает эффективный шаг по координатам, где градиенты часто велики, потому что знаменатель содержит накопленную сумму квадратов градиента. Это полезно при разных масштабах координат: направление автоматически нормируется по истории. Однако накопитель только растет, поэтому на длинных детерминированных запусках эффективный шаг монотонно уменьшается и метод может почти остановиться до достижения требуемой точности.

3.4 RMSProp

RMSProp заменяет полную сумму квадратов экспоненциальным средним:

$$s_t = \rho s_{t-1} + (1 - \rho) \nabla f(x_t)^2,$$

$$x_{t+1} = x_t - \alpha \frac{\nabla f(x_t)}{\sqrt{s_t} + \varepsilon_{\text{num}}}.$$

Промежуточный вывод. Экспоненциальное сглаживание предотвращает бесконечный рост накопителя, потому что старые квадраты градиента постепенно забываются с коэффициентом ρ . В отличие от AdaGrad, RMSProp масштабирует шаг по недавней, а не по всей накопленной истории. Поэтому метод лучше адаптируется, когда масштаб градиента меняется по мере движения к минимуму.

3.5 AdaDelta

AdaDelta использует экспоненциальные средние квадратов градиентов и квадратов обновлений:

$$E[g^2]_t = \rho E[g^2]_{t-1} + (1 - \rho) g_t^2,$$

$$\Delta x_t = - \frac{\sqrt{E[\Delta x^2]_{t-1} + \varepsilon_{\text{num}}}}{\sqrt{E[g^2]_t + \varepsilon_{\text{num}}}} g_t,$$

$$E[\Delta x^2]_t = \rho E[\Delta x^2]_{t-1} + (1 - \rho) \Delta x_t^2.$$

Промежуточный вывод. Метод не требует явного глобального шага, потому что масштаб обновления определяется отношением накопленного среднего квадратов прошлых шагов к среднему квадратов градиентов. Такая самонормировка делает AdaDelta устойчивым к выбору learning rate, но в рассматриваемых задачах приводит к консервативным обновлениям: если средний размер прошлых шагов мал, новые шаги также остаются малыми, и метод часто доходит до лимита итераций.

3.6 Adam

Adam объединяет Momentum и RMSProp:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t,$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2.$$

Используется коррекция смещения:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}.$$

Обновление:

$$x_{t+1} = x_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon_{\text{num}}}.$$

Промежуточный вывод. Adam выглядит самым сбалансированным из рассмотренных адаптивных методов: первый момент m_t сглаживает направление движения, а второй момент v_t нормирует шаг по координатам. Коррекция смещения нужна в начале оптимизации, когда накопители еще почти пустые и без поправки первые обновления были бы заметно искажены.

4 Структура программной реализации

Лабораторная работа построена поверх той же библиотеки `optimlib`, что и лабораторная №2. Ключевые элементы:

- `AdaptiveOptimizer` — базовый класс для методов с внутренним состоянием;
- `_reset_state()` — инициализация буферов перед каждым запуском;
- `_adaptive_direction(...)` — метод, в котором конкретный оптимизатор строит вектор обновления;
- `Momentum`, `Nesterov`, `AdaGrad`, `RMSProp`, `AdaDelta`, `Adam` — реализации методов;
- `OptimizationExperiment` — перебор функций, стартовых точек и сеток параметров;

- `plot_param_sensitivity` — построение line-графиков и heatmap чувствительности.

Сетки гиперпараметров задаются в `lab3/config.yaml`. Для Momentum и Nesterov перебираются α и β , для RMSProp — α и ρ , для AdaGrad — α , для AdaDelta — ρ , для Adam — α , β_1 и β_2 .

Промежуточный вывод. Общая архитектура позволила добавить шесть новых оптимизаторов без изменения кода функций и раннера. Различия методов сосредоточены в построении вектора обновления: базовый класс ведет цикл оптимизации, проверяет критерии останова и хранит историю, а конкретные методы держат свое состояние — скорость, накопитель квадратов градиента или моменты Adam. Это хорошо совпадает с математикой методов с памятью.

5 Результаты вычислительных экспериментов

5.1 Квадратичные функции

В таблице приведены лучшие наборы параметров по числу итераций на квадратичных функциях.

Таблица 1: Лучшие параметры на квадратичных функциях при $\varepsilon = 10^{-8}$

Функция	Метод	α	β/ρ	β_1	β_2	Итерации
f	Momentum	10^{-2}	0.85	—	—	182
f	Nesterov	10^{-2}	0.85	—	—	202
f	AdaGrad	10^{-3}	—	—	—	1500
f	RMSProp	10^{-2}	0.85	—	—	238
f	AdaDelta	—	0.50	—	—	1500
f	Adam	10^{-2}	—	0.70	0.90	289
g	Momentum	10^{-2}	0.85	—	—	260
g	Nesterov	10^{-2}	0.85	—	—	202
g	AdaGrad	10^{-3}	—	—	—	1500
g	RMSProp	10^{-2}	0.85	—	—	239
g	AdaDelta	—	0.50	—	—	1500
g	Adam	10^{-2}	—	0.70	0.95	362

Промежуточный вывод. На квадратичных функциях наиболее быстрыми оказались Momentum, Nesterov и RMSProp. У инерционных методов это объясняется накоплением направления: на квадратичной поверхности градиенты меняются достаточно предсказуемо, поэтому скорость помогает проходить пологие участки. RMSProp выигрывает за счет координатного масштабирования, полезного особенно на плохо обусловленной функции. AdaGrad и AdaDelta в выбранной сетке часто не достигали точности до лимита, потому что их механизмы уменьшения эффективного шага делали обновления слишком осторожными.

5.2 Чувствительность к параметрам

Для методов с двумя параметрами строились heatmap, где цвет отражает число итераций. Для методов с одним параметром строились line-графики.

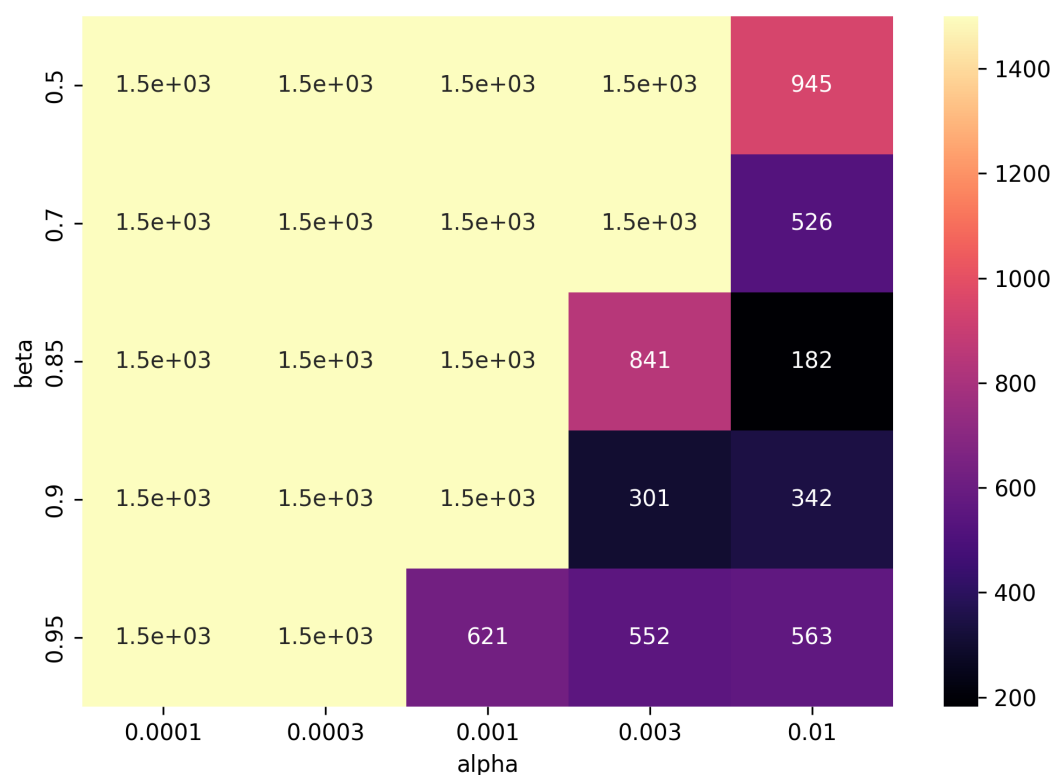


Рис. 1: Чувствительность Momentum к параметрам α и β

Комментарий к графику. Для Momentum важна пара “шаг–инерция”: большой β ускоряет движение только тогда, когда α не делает обновления слишком резкими. Поэтому на heatmap обычно видна область удачных сочетаний, а не один изолированный параметр.

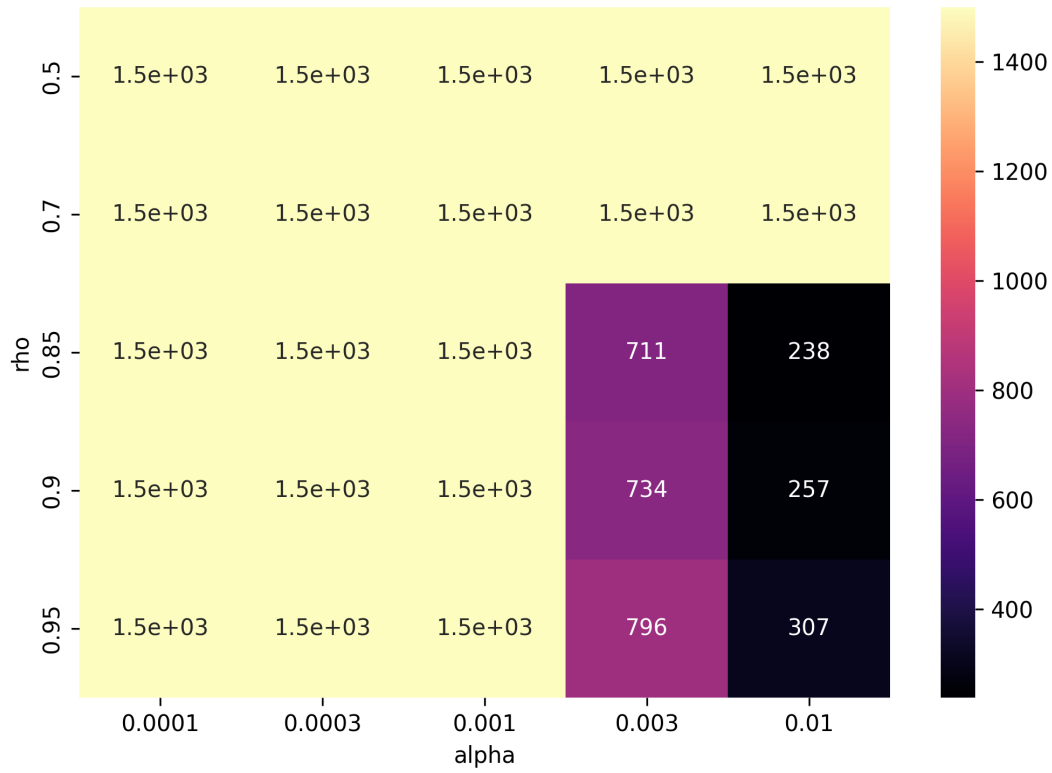


Рис. 2: Чувствительность RMSProp к параметрам α и ρ

Комментарий к графику. RMSProp масштабирует шаг через экспоненциальное среднее квадратов градиента. Если ρ слишком мал, масштаб быстро меняется от текущего градиента; если слишком велик, метод медленно реагирует на новую область поверхности.

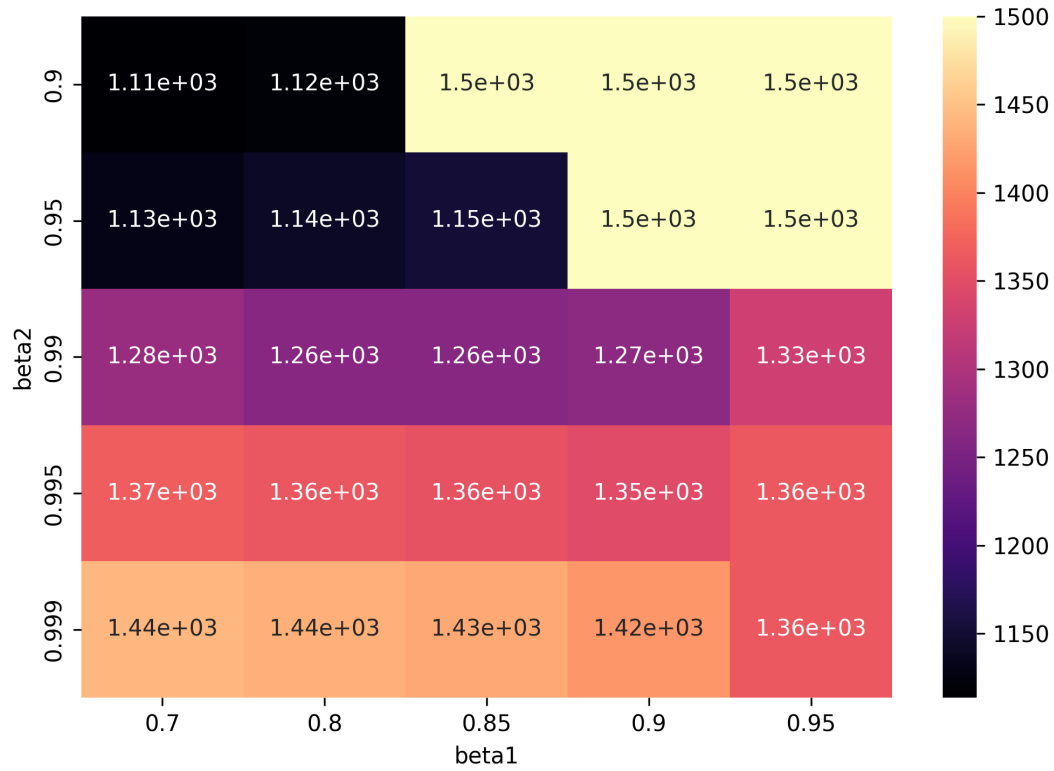


Рис. 3: Чувствительность Adam к параметрам β_1 и β_2

Комментарий к графику. У Adam первый момент отвечает за сглаженное направление, а второй — за координатный масштаб шага. Поэтому плохие сочетания β_1 и β_2 могут либо переусреднять направление, либо слишком медленно менять нормировку.

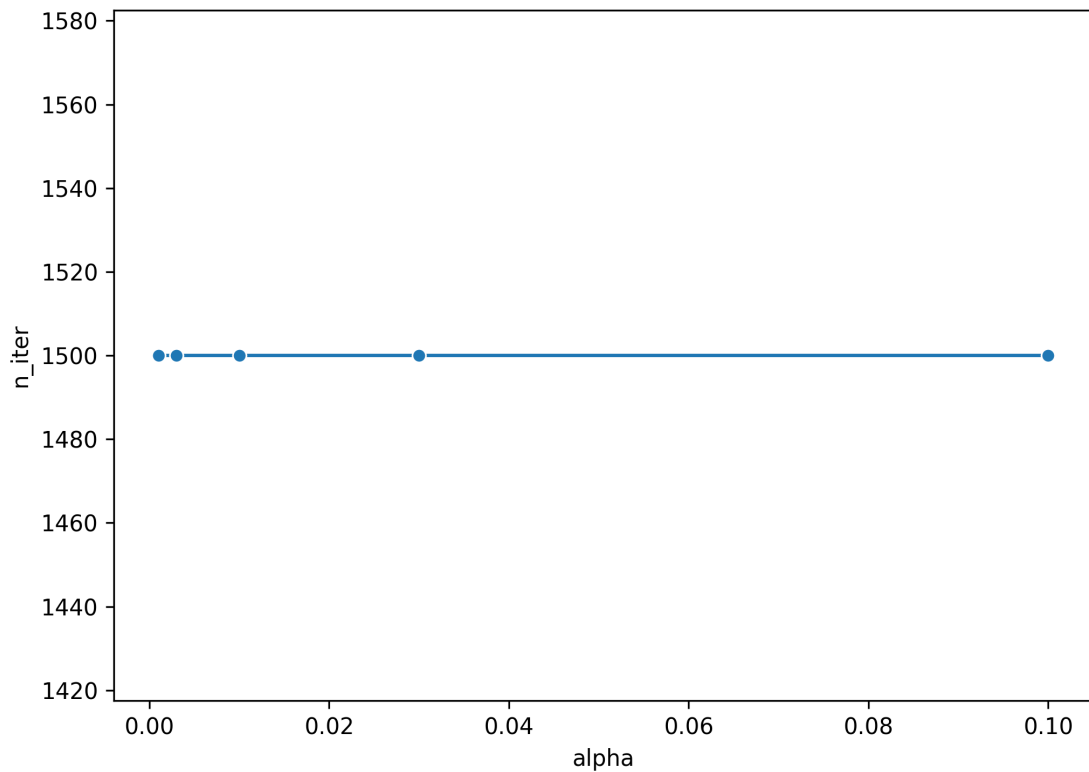


Рис. 4: Чувствительность AdaGrad к начальному шагу

Комментарий к графику. У AdaGrad накопитель квадратов градиента только растет, поэтому начальный шаг особенно важен. Слишком малый α быстро делает движение почти незаметным, а слишком большой может дать резкие первые обновления.

Промежуточный вывод. Для инерционных методов важна согласованность шага и коэффициента инерции: α задает силу текущего градиента, а β определяет, сколько прошлой скорости переносится в новый шаг. Если оба эффекта велики, обновление может стать слишком агрессивным и начать колебаться. Для адаптивных методов аналогичную роль играет сглаживание: малое сглаживание быстро реагирует на текущий градиент, а большое делает масштаб шага стабильнее, но медленнее приспосабливается к новой области поверхности.

5.3 Трактории на квадратичных функциях

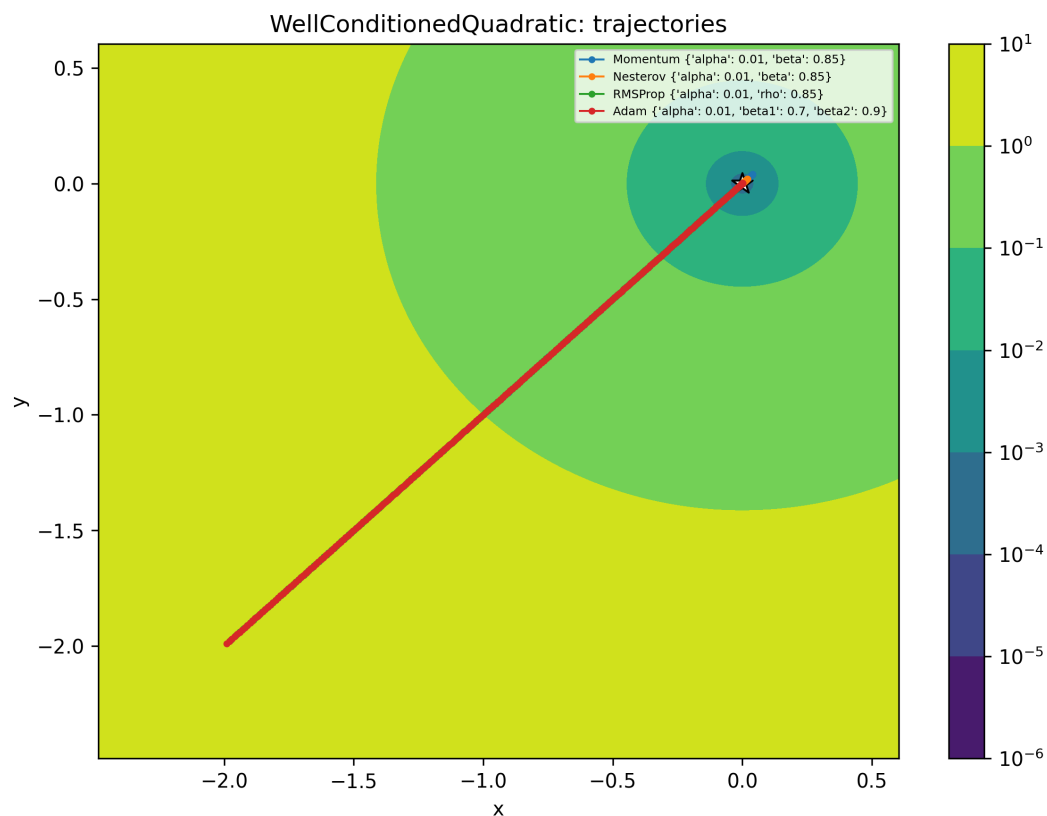


Рис. 5: Лучшие траектории методов на хорошо обусловленной квадратичной функции

Комментарий к графику. На хорошо обусловленной квадратичной функции направления градиента стабильны, поэтому методы с памятью используют прошлые шаги почти без вреда. Инерция здесь чаще ускоряет движение, чем мешает.

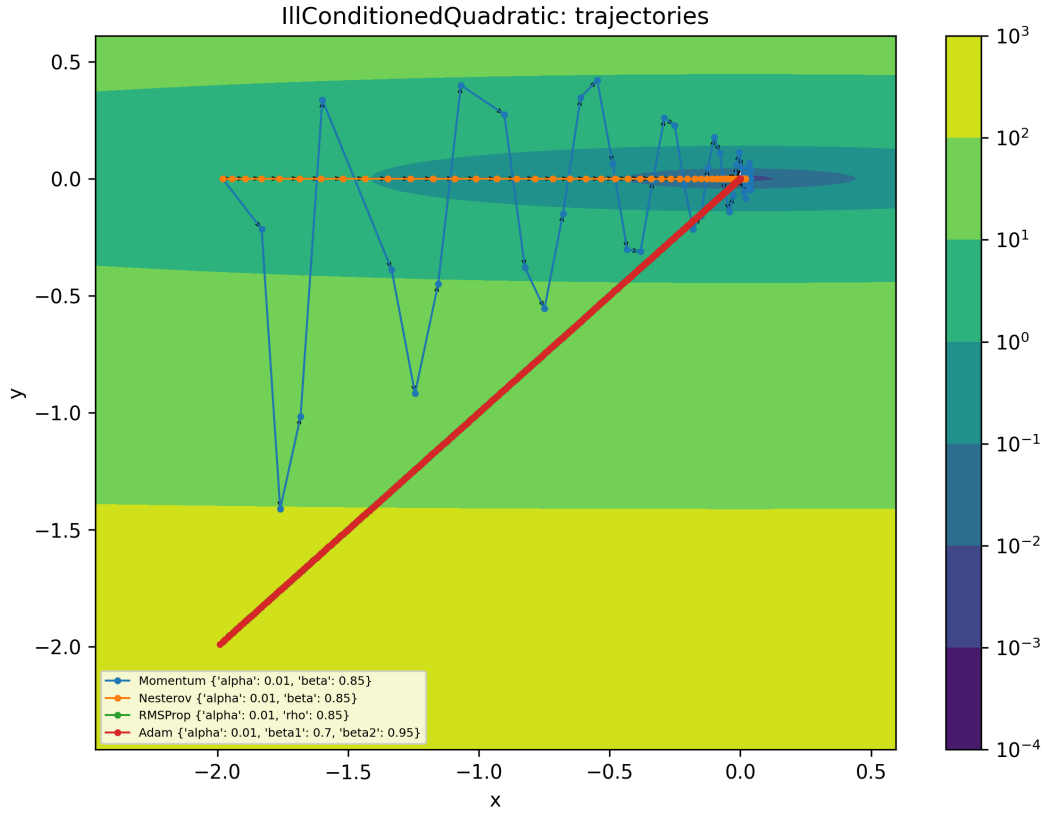


Рис. 6: Лучшие траектории методов на плохо обусловленной квадратичной функции

Комментарий к графику. На плохо обусловленной функции важна разная кривизна по координатам. RMSProp и Adam уменьшают влияние крутой координаты через нормировку, а Momentum и Nesterov помогают проходить вытянутую долину за счет накопленной скорости.

Промежуточный вывод. На хорошо обусловленной функции траектории в основном направлены к центру линий уровня, потому что масштаб по координатам одинаков и направление градиента хорошо согласовано с направлением к минимуму. На плохо обусловленной функции адаптивное масштабирование и инерция уменьшают влияние вытянутости разными способами: инерция накапливает движение вдоль устойчивого направления, а RMSProp и Adam уменьшают шаг по координатам с большими недавними градиентами. Поэтому они выигрывают у простого градиентного спуска с малым шагом.

5.4 Сложные функции

В таблице приведены лучшие результаты по значению функции среди исследованных параметров и стартовых точек.

Таблица 2: Лучшие запуски на сложных функциях

Функция	Метод	Лучший набор параметров	Итерации
Ackley	Momentum	$\alpha = 3 \cdot 10^{-4}, \beta = 0.95$	562
Ackley	Nesterov	$\alpha = 3 \cdot 10^{-3}, \beta = 0.50$	43
Ackley	AdaGrad	$\alpha = 10^{-2}$	81
Ackley	RMSProp	$\alpha = 10^{-4}, \rho = 0.90$	376
Ackley	AdaDelta	$\rho = 0.85$	175
Ackley	Adam	$\alpha = 3 \cdot 10^{-4}, \beta_1 = 0.70, \beta_2 = 0.95$	230
Rosenbrock	Momentum	$\alpha = 3 \cdot 10^{-3}, \beta = 0.90$	1277
Rosenbrock	Nesterov	$\alpha = 10^{-3}, \beta = 0.95$	1500
Rosenbrock	Adam	$\alpha = 10^{-2}, \beta_1 = 0.85, \beta_2 = 0.99$	1500
Himmelblau	Momentum	$\alpha = 3 \cdot 10^{-3}, \beta = 0.50$	63
Himmelblau	Nesterov	$\alpha = 10^{-2}, \beta = 0.50$	24
Himmelblau	Adam	$\alpha = 10^{-3}, \beta_1 = 0.85, \beta_2 = 0.99$	1339

Промежуточный вывод. На функции Химмельблау методы обычно находили один из глобальных минимумов, потому что после попадания в область притяжения градиентные направления достаточно уверенно ведут к ближайшему минимуму. На функции Розенброка ситуация сложнее: узкая изогнутая долина требует не только правильного масштаба шага, но и постоянного изменения направления вдоль кривой. Инерция может ускорять движение, но также может уносить поперек долины; адаптивные масштабы помогают с длиной шага, но не устраняют полностью проблему направления.

5.5 Травектории на сложных функциях

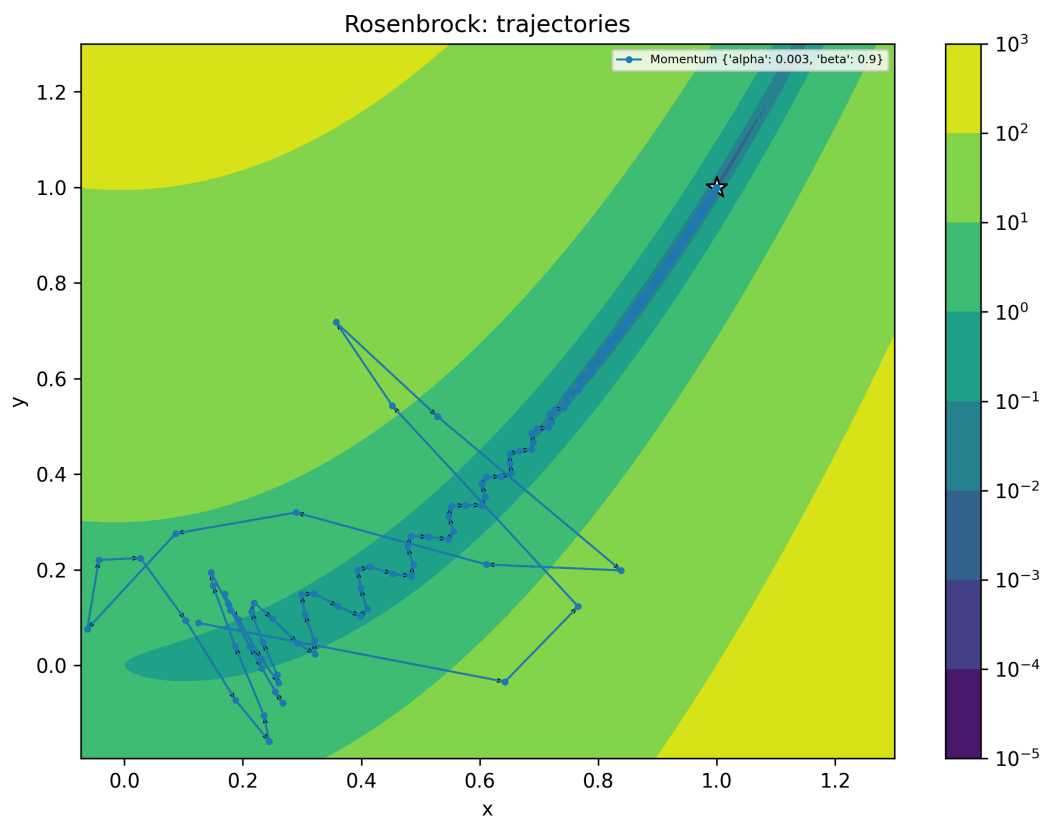


Рис. 7: Лучшие траектории методов на функции Розенброка

Комментарий к графику. У Розенброка узкая изогнутая долина, поэтому инерция может как ускорять движение вдоль долины, так и уносить траекторию поперек нее. Адаптивный масштаб шага помогает, но не заменяет информацию о кривизне.

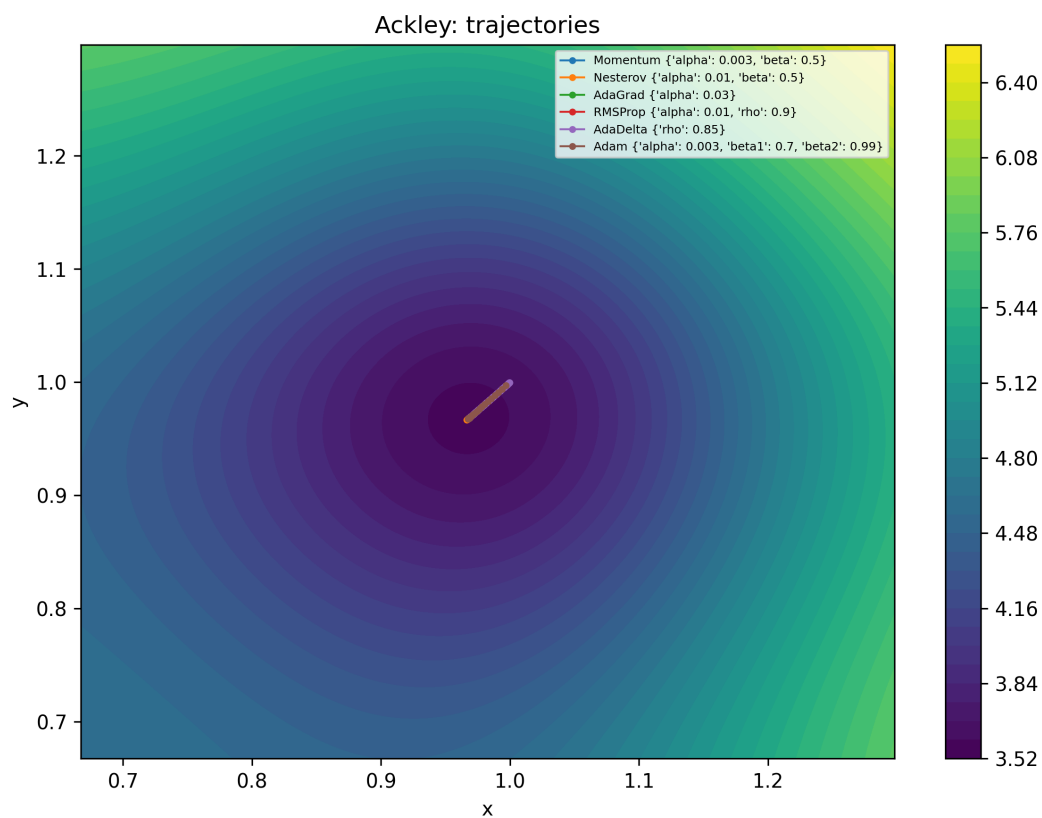


Рис. 8: Лучшие траектории методов на функции Экли

Комментарий к графику. У Экли локальные колебания поверхности меняют направление градиента. Методы с накоплением истории сглаживают часть этих колебаний, но при неудачных параметрах могут сохранять движение в уже неактуальном направлении.

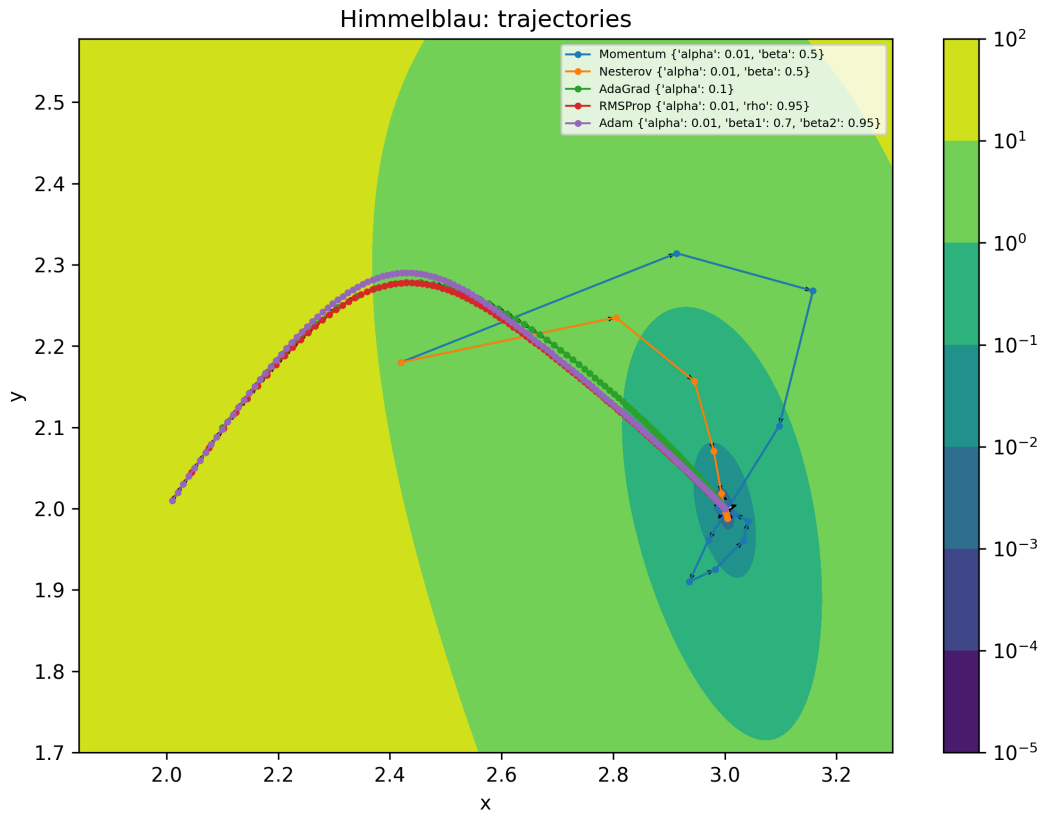


Рис. 9: Лучшие траектории методов на функции Химмельблау

Комментарий к графику. На Химмельблау траектория во многом определяется областью притяжения стартовой точки. После попадания в одну из таких областей методы различаются главным образом скоростью подхода к ближайшему минимуму.

Промежуточный вывод. Визуализация траекторий особенно важна для сложных функций: одинаковое итоговое число итераций может соответствовать совершенно разному поведению. У инерционных методов на графиках видны более длинные и сглаженные перемещения, поскольку шаг зависит от прошлой скорости. У адаптивных методов траектория может быть менее прямой, зато координатное масштабирование уменьшает слишком резкие движения в направлениях с большими градиентами. Поэтому график траектории дополняет таблицу и показывает, как именно внутренняя логика метода влияет на путь к минимуму.

6 Сводная таблица эксперимента

Полная сводка всех 1470 запусков сохраняется в файле `../outputs/lab3/summary.csv`

Ниже приведена печатная версия этой сводки: строки разбиты на несколько масштабируемых блоков, чтобы таблица оставалась в пределах страницы.

Таблица 3: Full Lab 3 experiment table: rows 1–32 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.5}	no	2.196	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.7}	no	1.472	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.85}	no	0.541	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.9}	no	0.197	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.95}	no	0.008367	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.5}	no	0.6609	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.7}	no	0.1984	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.85}	no	0.009408	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.9}	no	3.99e-04	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.95}	no	5.09e-09	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.5}	no	0.009776	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.7}	no	1.68e-04	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.85}	no	3.66e-09	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.9}	no	1.35e-14	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.95}	yes	4.82e-19	621	622	622
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.5}	no	5.23e-08	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.7}	no	1.60e-13	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.85}	yes	4.84e-17	841	842	842
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.9}	yes	4.78e-19	301	302	302
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.95}	yes	2.37e-18	552	553	553
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.5}	yes	4.80e-17	945	946	946
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.7}	yes	4.70e-17	526	527	527
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.85}	yes	3.32e-17	182	183	183
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.9}	yes	4.25e-17	342	343	343
WellConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.95}	yes	4.05e-17	563	564	564
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.5}	no	2.196	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.7}	no	1.473	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.85}	no	0.5416	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.9}	no	0.1976	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.95}	no	0.008473	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.5}	no	0.6613	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.7}	no	0.1988	1500	1501	3001

Таблица 4: Full Lab 3 experiment table: rows 33–64 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.85}	no	0.009508	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.9}	no	4.09e-04	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.95}	no	5.94e-09	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.5}	no	0.009836	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.7}	no	1.72e-04	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.85}	no	4.16e-09	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.9}	no	1.96e-14	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.95}	yes	3.73e-17	643	644	1287
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.5}	no	5.53e-08	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.7}	no	2.01e-13	1500	1501	3001
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.85}	yes	4.79e-17	861	862	1723
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.9}	yes	4.83e-17	367	368	735
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.95}	yes	3.37e-17	625	626	1251
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.5}	yes	4.83e-17	955	956	1911
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.7}	yes	4.89e-17	541	542	1083
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.85}	yes	4.07e-17	202	203	405
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.9}	yes	1.78e-18	246	247	493
WellConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.95}	yes	3.72e-17	473	474	947
WellConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.001}	no	3.704	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.003}	no	3.156	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.01}	no	1.68	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.03}	no	0.1182	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.1}	no	5.15e-12	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.5}	no	3.422	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.7}	no	3.422	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.85}	no	3.421	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.9}	no	3.419	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.95}	no	3.415	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.5}	no	2.402	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.7}	no	2.401	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.85}	no	2.399	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.9}	no	2.396	1500	1501	1501

Таблица 5: Full Lab 3 experiment table: rows 65–96 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.95}	no	2.387	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.5}	no	0.2507	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.7}	no	0.2513	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.85}	no	0.2524	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.9}	no	0.2532	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.95}	no	0.2545	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.5}	no	2.25e-06	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.7}	no	2.25e-06	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.85}	yes	2.15e-18	711	712	712
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.9}	yes	1.07e-17	734	735	735
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.95}	yes	2.76e-17	796	797	797
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.5}	no	2.50e-05	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.7}	no	2.50e-05	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.85}	yes	3.61e-19	238	239	239
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.9}	yes	1.19e-17	257	258	258
WellConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.95}	yes	2.30e-17	307	308	308
WellConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.5}	no	6.95e-04	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.7}	no	0.133	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.85}	no	0.5441	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.9}	no	0.802	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.95}	no	1.114	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	3.423	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	3.423	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	3.425	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	3.427	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	3.431	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	3.423	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	3.423	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	3.425	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	3.427	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	3.431	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	3.423	1500	1501	1501

Таблица 6: Full Lab 3 experiment table: rows 97–128 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	3.423	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	3.425	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	3.427	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	3.431	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	3.423	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	3.423	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	3.425	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	3.427	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	3.431	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	3.422	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	3.423	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	3.425	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	3.426	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	3.431	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	2.404	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	2.406	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	2.423	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	2.438	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	2.472	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	2.404	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	2.406	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	2.422	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	2.438	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	2.472	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	2.403	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	2.406	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	2.422	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	2.437	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	2.471	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	2.403	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	2.405	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	2.421	1500	1501	1501

Таблица 7: Full Lab 3 experiment table: rows 129–160 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	2.437	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	2.471	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	2.4	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	2.403	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	2.419	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	2.434	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	2.469	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	0.2591	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	0.2724	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	0.3637	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	0.4454	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	0.6025	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	0.2568	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	0.2702	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	0.3617	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	0.4437	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	0.601	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	0.2546	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	0.268	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	0.3598	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	0.4419	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	0.5996	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	0.2501	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	0.2635	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	0.3559	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	0.4384	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	0.5967	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	0.2365	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	0.2503	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	0.3443	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	0.428	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	0.588	1500	1501	1501

Таблица 8: Full Lab 3 experiment table: rows 161–192 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	1.33e-18	779	780	780
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	1.69e-17	818	819	819
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	4.61e-17	1266	1267	1267
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	1.82e-09	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	6.80e-04	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	1.26e-17	778	779	779
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	8.73e-18	831	832	832
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	4.38e-17	1234	1235	1235
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	9.31e-10	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	6.40e-04	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	2.26e-08	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	yes	1.08e-17	852	853	853
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	yes	4.30e-17	1229	1230	1230
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	4.26e-10	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	6.02e-04	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	8.16e-09	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	1.29e-08	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	yes	7.72e-18	1227	1228	1228
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	4.94e-11	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	5.30e-04	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	3.67e-09	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	1.14e-09	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	yes	3.99e-17	1360	1361	1361
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	5.38e-14	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	3.49e-04	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	yes	4.09e-17	289	290	290
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	yes	2.82e-17	330	331	331
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	yes	3.99e-17	627	628	628
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	yes	4.47e-17	840	841	841
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	yes	4.97e-17	1207	1208	1208
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	yes	2.69e-17	319	320	320
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	yes	2.82e-17	370	371	371

Таблица 9: Full Lab 3 experiment table: rows 193–224 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	yes	3.81e-17	587	588	588
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	yes	5.00e-17	812	813	813
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	yes	4.97e-17	1178	1179	1179
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	no	2.65e-09	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	4.99e-17	406	407	407
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	3.75e-17	583	584	584
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	4.33e-17	778	779	779
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	4.95e-17	1147	1148	1148
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	no	4.76e-08	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	no	4.20e-08	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	yes	3.67e-17	618	619	619
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	yes	4.70e-17	738	739	739
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	yes	4.94e-17	1075	1076	1076
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	3.47e-08	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	1.32e-08	1500	1501	1501
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	yes	3.96e-17	785	786	786
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	yes	3.75e-17	820	821	821
WellConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	yes	1.18e-17	815	816	816
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.5}	no	1.098	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.7}	no	0.7362	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.85}	no	0.2705	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.9}	no	0.09852	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.95}	no	0.004184	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.5}	no	0.3305	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.7}	no	0.09919	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.85}	no	0.004704	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.9}	no	1.99e-04	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.95}	no	2.54e-09	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.5}	no	0.004888	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.7}	no	8.38e-05	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.85}	no	1.83e-09	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.9}	no	6.73e-15	1500	1501	1501

Таблица 10: Full Lab 3 experiment table: rows 225–256 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.001, 'beta': 0.95}	yes	1.06e-18	808	809	809
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.5}	no	2.62e-08	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.7}	no	7.99e-14	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.85}	yes	4.87e-17	826	827	827
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.9}	yes	2.15e-17	390	391	391
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.003, 'beta': 0.95}	yes	1.00e-17	780	781	781
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.5}	yes	4.85e-17	928	929	929
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.7}	yes	4.95e-17	516	517	517
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.85}	yes	1.85e-18	260	261	261
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.9}	yes	2.58e-18	392	393	393
IllConditionedQuadratic	[-2.0, -2.0]	Momentum	{'alpha': 0.01, 'beta': 0.95}	yes	8.47e-19	791	792	792
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.5}	no	1.098	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.7}	no	0.7364	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.85}	no	0.2708	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.9}	no	0.09879	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.95}	no	0.004236	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.5}	no	0.3306	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.7}	no	0.0994	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.85}	no	0.004754	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.9}	no	2.05e-04	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.95}	no	2.97e-09	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.5}	no	0.004918	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.7}	no	8.58e-05	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.85}	no	2.08e-09	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.9}	no	9.79e-15	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.95}	yes	4.52e-17	642	643	1285
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.5}	no	2.76e-08	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.7}	no	1.00e-13	1500	1501	3001
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.85}	yes	4.97e-17	845	846	1691
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.9}	yes	4.76e-17	363	364	727
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.95}	yes	1.68e-17	625	626	1251
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.5}	yes	4.84e-17	938	939	1877

Таблица 11: Full Lab 3 experiment table: rows 257–288 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.7}	yes	4.69e-17	532	533	1065
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.85}	yes	2.04e-17	202	203	405
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.9}	yes	8.88e-19	246	247	493
IllConditionedQuadratic	[-2.0, -2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.95}	yes	1.86e-17	473	474	947
IllConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.001}	no	187	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.003}	no	159.4	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.01}	no	84.84	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.03}	no	5.969	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	AdaGrad	{'alpha': 0.1}	no	2.60e-10	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.5}	no	172.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.7}	no	172.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.85}	no	172.7	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.9}	no	172.7	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.95}	no	172.5	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.5}	no	121.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.7}	no	121.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.85}	no	121.1	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.9}	no	121	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.95}	no	120.6	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.5}	no	12.66	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.7}	no	12.69	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.85}	no	12.75	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.9}	no	12.78	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.95}	no	12.85	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.003, 'rho': 0.5}	no	1.14e-04	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.003, 'rho': 0.7}	no	1.14e-04	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.003, 'rho': 0.85}	yes	6.49e-20	712	713	713
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.003, 'rho': 0.9}	yes	1.08e-19	737	738	738
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.003, 'rho': 0.95}	yes	1.55e-19	804	805	805
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.01, 'rho': 0.5}	no	0.001262	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.01, 'rho': 0.7}	no	0.001262	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	RMSPProp	{'alpha': 0.01, 'rho': 0.85}	yes	1.53e-20	239	240	240

Таблица 12: Full Lab 3 experiment table: rows 289–320 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
IllConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.9}	yes	4.30e-20	260	261	261
IllConditionedQuadratic	[-2.0, -2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.95}	yes	2.27e-19	314	315	315
IllConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.5}	no	0.03508	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.7}	no	6.716	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.85}	no	27.48	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.9}	no	40.5	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	AdaDelta	{'rho': 0.95}	no	56.26	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	172.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	172.9	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	173	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	173.1	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	173.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	172.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	172.9	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	173	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	173.1	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	173.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	172.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	172.9	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	173	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	173.1	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	173.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	172.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	172.9	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	173	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	173	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	172.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	172.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	172.9	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	173	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	173.3	1500	1501	1501

Таблица 13: Full Lab 3 experiment table: rows 321–352 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	121.4	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	121.5	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	122.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	123.1	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	124.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	121.4	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	121.5	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	122.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	123.1	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	124.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	121.4	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	121.5	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	122.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	123.1	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	124.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	121.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	121.4	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	122.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	123	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	124.8	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	121.2	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	121.3	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	122.2	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	122.9	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	124.7	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	13.09	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	13.76	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	18.37	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	22.49	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	30.43	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	12.97	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	13.64	1500	1501	1501

Таблица 14: Full Lab 3 experiment table: rows 353–384 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	no	18.27	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	no	22.4	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	no	30.35	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	no	12.86	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	no	13.53	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	no	18.17	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	no	22.32	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	no	30.28	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	12.63	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	13.31	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	17.97	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	22.14	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	30.13	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	11.94	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	12.64	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	17.39	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	21.62	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	29.69	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	3.78e-07	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	7.30e-21	833	834	834
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	3.73e-19	1298	1299	1299
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	9.21e-08	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	0.03434	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	2.60e-06	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	9.09e-20	876	877	877
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	4.67e-19	1242	1243	1243
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	4.70e-08	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	0.03232	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	1.16e-06	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	yes	2.69e-20	916	917	917
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	yes	2.35e-19	1237	1238	1238
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	2.15e-08	1500	1501	1501

Таблица 15: Full Lab 3 experiment table: rows 385–416 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	0.03039	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	4.32e-07	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	6.92e-09	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	yes	3.99e-20	1306	1307	1307
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	2.50e-09	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	0.02676	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	1.84e-07	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	5.34e-08	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	yes	1.63e-20	1487	1488	1488
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	2.72e-12	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	0.01763	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	no	8.03e-06	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	2.17e-19	362	363	363
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	3.83e-19	654	655	655
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	4.48e-19	900	901	901
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	4.96e-19	1351	1352	1352
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	no	2.29e-05	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	2.29e-19	411	412	412
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	3.93e-19	628	629	629
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	4.73e-19	864	865	865
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	4.88e-19	1316	1317	1317
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	no	7.85e-08	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	4.57e-19	479	480	480
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	8.44e-22	624	625	625
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	3.32e-19	807	808	808
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	4.76e-19	1278	1279	1279
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	2.29e-06	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	4.73e-06	1500	1501	1501
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	yes	2.63e-19	656	657	657
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	yes	2.91e-19	809	810	810
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	yes	4.71e-19	1185	1186	1186
IIIConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	1.78e-06	1500	1501	1501

Таблица 16: Full Lab 3 experiment table: rows 417–448 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	6.24e-07	1500	1501	1501
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	yes	4.01e-19	916	917	917
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	yes	2.90e-19	997	998	998
IllConditionedQuadratic	[-2.0, -2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	yes	2.08e-19	964	965	965
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.5}	no	3.116	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.7}	no	2.295	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.85}	no	0.4423	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.9}	no	0.152	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.95}	no	0.01993	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.5}	no	0.6203	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.7}	no	0.159	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.85}	no	0.02322	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.9}	no	0.004975	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.95}	no	5.38e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.5}	no	0.02437	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.7}	no	0.003511	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.85}	no	4.09e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.9}	no	3.86e-07	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.95}	no	3.79e-14	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.5}	no	7.51e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.7}	no	5.28e-07	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.85}	no	1.23e-12	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.9}	yes	1.24e-16	1277	1278	1278
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.95}	—	—	—	—	—
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.5}	—	—	—	—	—
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.7}	—	—	—	—	—
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.85}	—	—	—	—	—
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.9}	—	—	—	—	—
Rosenbrock	[-1.2, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.95}	—	—	—	—	—
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.5}	no	3.117	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.7}	no	2.301	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.85}	no	0.4518	1500	1501	3001

Таблица 17: Full Lab 3 experiment table: rows 449–480 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.9}	no	0.1573	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.95}	no	0.02238	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.5}	no	0.6233	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.7}	no	0.1607	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.85}	no	0.02416	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.9}	no	0.00544	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.95}	no	7.85e-05	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.5}	no	0.02437	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.7}	no	0.003562	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.85}	no	4.63e-05	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.9}	no	5.41e-07	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.95}	no	1.91e-15	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.5}	no	0.1245	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.7}	no	0.1138	1500	1501	3001
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.85}	–	–	–	–	–
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.9}	–	–	–	–	–
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.95}	–	–	–	–	–
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.5}	–	–	–	–	–
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.7}	–	–	–	–	–
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.85}	–	–	–	–	–
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.9}	–	–	–	–	–
Rosenbrock	[-1.2, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.95}	–	–	–	–	–
Rosenbrock	[-1.2, 1.0]	AdaGrad	{'alpha': 0.001}	no	9.296	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	AdaGrad	{'alpha': 0.003}	no	4.302	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	AdaGrad	{'alpha': 0.01}	no	4.166	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	AdaGrad	{'alpha': 0.03}	no	3.557	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	AdaGrad	{'alpha': 0.1}	no	0.6649	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.5}	no	4.226	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.7}	no	4.226	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.85}	no	4.227	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.9}	no	4.229	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.95}	no	4.237	1500	1501	1501

Таблица 18: Full Lab 3 experiment table: rows 481–512 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.5}	no	3.62	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.7}	no	3.622	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.85}	no	3.627	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.9}	no	3.633	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.95}	no	3.654	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.5}	no	1.013	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.7}	no	1.013	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.85}	no	1.016	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.9}	no	1.023	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.95}	no	1.053	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.5}	no	0.01955	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.7}	no	0.01507	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.85}	no	0.01318	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.9}	no	0.01294	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.95}	no	0.01352	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.5}	no	0.03197	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.7}	no	0.02754	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.85}	no	0.02629	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.9}	no	0.02571	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.95}	no	0.02563	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	AdaDelta	{'rho': 0.5}	no	0.06765	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	AdaDelta	{'rho': 0.7}	no	0.1568	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	AdaDelta	{'rho': 0.85}	no	1.049	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	AdaDelta	{'rho': 0.9}	no	2.051	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	AdaDelta	{'rho': 0.95}	no	3.021	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	4.231	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	4.241	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	4.354	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	4.594	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	5.212	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	4.231	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	4.24	1500	1501	1501

Таблица 19: Full Lab 3 experiment table: rows 513–544 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	4.35	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	4.588	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	5.205	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	4.231	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	4.239	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	4.346	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	4.583	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	5.199	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	4.231	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	4.237	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	4.338	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	4.571	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	5.186	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	4.23	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	4.236	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	4.316	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	4.539	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	5.149	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	3.637	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	3.665	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	3.946	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	4.182	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	4.243	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	3.636	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	3.664	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	3.945	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	4.182	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	4.243	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	3.636	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	3.663	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	3.944	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	4.181	1500	1501	1501

Таблица 20: Full Lab 3 experiment table: rows 545–576 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	4.243	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	3.635	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	3.66	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	3.941	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	4.18	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	4.243	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	3.637	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	3.657	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	3.934	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	4.177	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	4.242	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	1.03	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	1.064	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	2.344	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	3.55	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	4.087	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	1.043	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	1.074	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	2.339	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	3.545	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	4.085	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	1.061	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	1.09	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	2.334	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	3.539	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	4.084	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	1.091	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	1.133	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	2.326	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	3.53	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	4.082	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	1.41	1500	1501	1501

Таблица 21: Full Lab 3 experiment table: rows 577–608 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2': ...}	no	1.247	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2': ...}	no	2.309	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2': ...}	no	3.507	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2': ...}	no	4.075	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2': ...}	no	5.14e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2': ...}	no	1.85e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2': ...}	no	0.01699	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2': ...}	no	0.7842	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2': ...}	no	3.267	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2': ...}	no	4.54e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2': ...}	no	9.88e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2': ...}	no	0.01553	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2': ...}	no	0.7603	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2': ...}	no	3.251	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...}	no	2.52e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...}	no	3.38e-07	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...}	no	0.0143	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...}	no	0.741	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...}	no	3.238	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...}	no	1.35e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...}	no	7.55e-06	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...}	no	0.01236	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...}	no	0.7142	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...}	no	3.22	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...}	no	3.44e-06	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...}	no	3.71e-07	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...}	no	0.009333	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...}	no	0.6984	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...}	no	3.209	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	4.43e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	4.99e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	5.82e-06	1500	1501	1501

Таблица 22: Full Lab 3 experiment table: rows 609–640 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	0.03442	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	0.2165	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	2.29e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	3.98e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	4.14e-07	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	4.42e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	0.2087	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	no	2.25e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	no	3.26e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	no	1.56e-08	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	no	4.24e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	no	0.2086	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	no	4.32e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	no	1.03e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	no	3.33e-06	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	no	4.61e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	no	0.2211	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	no	5.30e-05	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	no	8.67e-06	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	no	5.10e-08	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	no	7.97e-04	1500	1501	1501
Rosenbrock	[-1.2, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	no	0.2922	1500	1501	1501
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.5}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.7}	yes	3.574	1112	1113	1113
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.85}	yes	3.574	429	430	430
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.9}	yes	3.574	238	239	239
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0001, 'beta': 0.95}	yes	3.574	597	598	598
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.5}	yes	3.574	619	620	620
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.7}	yes	3.574	327	328	328
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.85}	yes	3.574	184	185	185
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.9}	yes	3.574	322	323	323
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.0003, 'beta': 0.95}	yes	3.574	562	563	563

Таблица 23: Full Lab 3 experiment table: rows 641–672 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.5}	yes	3.574	162	163	163
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.7}	yes	3.574	90	91	91
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.85}	yes	3.574	211	212	212
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.9}	yes	3.574	319	320	320
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.001, 'beta': 0.95}	yes	3.574	583	584	584
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.5}	yes	3.574	50	51	51
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.7}	yes	3.574	95	96	96
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.85}	yes	3.574	203	204	204
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.9}	yes	3.574	323	324	324
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.003, 'beta': 0.95}	yes	3.574	639	640	640
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.5}	yes	3.574	54	55	55
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.7}	yes	3.574	104	105	105
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.85}	yes	3.574	209	210	210
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.9}	yes	3.574	314	315	315
Ackley	[1.0, 1.0]	Momentum	{'alpha': 0.01, 'beta': 0.95}	yes	3.574	465	466	466
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.5}	no	3.574	1500	1501	3001
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.7}	yes	3.574	1127	1128	2255
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.85}	yes	3.574	454	455	909
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.9}	yes	3.574	318	319	637
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.95}	yes	3.574	558	559	1117
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.5}	yes	3.574	630	631	1261
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.7}	yes	3.574	344	345	689
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.85}	yes	3.574	196	197	393
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.9}	yes	3.574	275	276	551
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.95}	yes	3.574	422	423	845
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.5}	yes	3.574	174	175	349
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.7}	yes	3.574	86	87	173
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.85}	yes	3.574	160	161	321
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.9}	yes	3.574	196	197	393
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.001, 'beta': 0.95}	yes	3.574	333	334	667
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.5}	yes	3.574	43	44	87
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.7}	yes	3.574	68	69	137

Таблица 24: Full Lab 3 experiment table: rows 673–704 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.85}	yes	3.574	94	95	189
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.9}	yes	3.574	125	126	251
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.003, 'beta': 0.95}	yes	3.574	164	165	329
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.5}	yes	3.574	28	29	57
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.7}	yes	3.574	35	36	71
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.85}	yes	3.574	43	44	87
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.9}	yes	3.574	47	48	95
Ackley	[1.0, 1.0]	Nesterov	{'alpha': 0.01, 'beta': 0.95}	yes	3.574	51	52	103
Ackley	[1.0, 1.0]	AdaGrad	{'alpha': 0.001}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	AdaGrad	{'alpha': 0.003}	yes	3.574	670	671	671
Ackley	[1.0, 1.0]	AdaGrad	{'alpha': 0.01}	yes	3.574	81	82	82
Ackley	[1.0, 1.0]	AdaGrad	{'alpha': 0.03}	yes	3.574	8	9	9
Ackley	[1.0, 1.0]	AdaGrad	{'alpha': 0.1}	yes	3.574	36	37	37
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.5}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.7}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.85}	yes	3.574	356	357	357
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.9}	yes	3.574	376	377	377
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.95}	yes	3.574	432	433	433
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.5}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.7}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.85}	yes	3.574	139	140	140
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.9}	yes	3.574	157	158	158
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.95}	yes	3.574	199	200	200
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.5}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.7}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.85}	yes	3.574	58	59	59
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.9}	yes	3.574	69	70	70
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.001, 'rho': 0.95}	yes	3.574	87	88	88
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.5}	no	3.575	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.7}	no	3.575	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.85}	yes	3.574	27	28	28
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.9}	yes	3.574	30	31	31

Таблица 25: Full Lab 3 experiment table: rows 705–736 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.003, 'rho': 0.95}	yes	3.574	29	30	30
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.5}	no	3.576	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.7}	no	3.576	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.85}	no	3.576	1500	1501	1501
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.9}	yes	3.574	6	7	7
Ackley	[1.0, 1.0]	RMSProp	{'alpha': 0.01, 'rho': 0.95}	no	3.576	1500	1501	1501
Ackley	[1.0, 1.0]	AdaDelta	{'rho': 0.5}	no	3.575	1500	1501	1501
Ackley	[1.0, 1.0]	AdaDelta	{'rho': 0.7}	no	3.575	1500	1501	1501
Ackley	[1.0, 1.0]	AdaDelta	{'rho': 0.85}	yes	3.574	175	176	176
Ackley	[1.0, 1.0]	AdaDelta	{'rho': 0.9}	yes	3.574	202	203	203
Ackley	[1.0, 1.0]	AdaDelta	{'rho': 0.95}	yes	3.574	256	257	257
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	yes	3.574	402	403	403
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	yes	3.574	455	456	456
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	yes	3.574	823	824	824
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	yes	3.574	1136	1137	1137
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	yes	3.574	476	477	477
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	yes	3.574	789	790	790
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	yes	3.574	1112	1113	1113
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	yes	3.574	537	538	538
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	yes	3.574	785	786	786
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	yes	3.574	1084	1085	1085
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	yes	3.574	609	610	610
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	yes	3.574	782	783	783
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	yes	3.574	994	995	995
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	3.574	1500	1501	1501

Таблица 26: Full Lab 3 experiment table: rows 737–768 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	yes	3.574	919	920	920
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	yes	3.574	1071	1072	1072
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	yes	3.574	1468	1469	1469
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	yes	3.574	203	204	204
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	yes	3.574	230	231	231
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	yes	3.574	416	417	417
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	yes	3.574	524	525	525
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	yes	3.574	660	661	661
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	yes	3.574	252	253	253
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	yes	3.574	365	366	366
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	yes	3.574	489	490	490
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	yes	3.574	626	627	627
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	yes	3.574	311	312	312
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	yes	3.574	362	363	363
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	yes	3.574	428	429	429
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	yes	3.574	586	587	587
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	yes	3.574	399	400	400
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	yes	3.574	447	448	448
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	yes	3.574	512	513	513
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	yes	3.574	652	653	653
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	yes	3.574	658	659	659
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	yes	3.574	591	592	592
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	yes	3.574	98	99	99
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	yes	3.574	132	133	133
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	yes	3.574	134	135	135

Таблица 27: Full Lab 3 experiment table: rows 769–800 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2':...	yes	3.574	151	152	152
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2':...	yes	3.574	164	165	165
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	yes	3.574	172	173	173
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	yes	3.574	168	169	169
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	yes	3.574	181	182	182
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	yes	3.574	171	172	172
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	yes	3.574	247	248	248
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	yes	3.574	220	221	221
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	yes	3.574	219	220	220
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	yes	3.574	210	211	211
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	yes	3.574	336	337	337
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	yes	3.574	299	300	300
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	yes	3.574	299	300	300
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	yes	3.574	554	555	555
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	yes	3.574	651	652	652
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	yes	3.574	641	642	642
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	3.574	102	103	103
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	3.574	90	91	91
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	3.574	105	106	106
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	3.574	100	101	101
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	3.574	155	156	156
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	3.574	154	155	155
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	3.574	162	163	163
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	3.574	160	161	161

Таблица 28: Full Lab 3 experiment table: rows 801–832 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...	yes	3.574	217	218	218
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...	yes	3.574	180	181	181
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2': ...	yes	3.574	188	189	189
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...	yes	3.574	307	308	308
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...	yes	3.574	312	313	313
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2': ...	yes	3.574	326	327	327
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...	yes	3.574	678	679	679
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...	yes	3.574	632	633	633
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...	yes	3.574	609	610	610
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	no	3.575	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	3.574	102	103	103
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	3.574	99	100	100
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	3.574	98	99	99
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	3.574	163	164	164
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	3.574	157	158	158
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	3.574	159	160	160
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...	yes	3.574	202	203	203
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...	yes	3.574	203	204	204
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...	yes	3.574	166	167	167
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	3.574	1500	1501	1501

Таблица 29: Full Lab 3 experiment table: rows 833–864 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	yes	3.574	339	340	340
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	yes	3.574	311	312	312
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	yes	3.574	275	276	276
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	no	3.574	1500	1501	1501
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	yes	3.574	673	674	674
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	yes	3.574	622	623	623
Ackley	[1.0, 1.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	yes	3.574	667	668	668
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.5}	no	4.98e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.7}	no	1.04e-11	1500	1501	1501
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.85}	yes	1.88e-18	1083	1084	1084
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.9}	yes	1.81e-18	495	496	496
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.95}	yes	5.63e-19	841	842	842
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.5}	yes	1.90e-18	1325	1326	1326
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.7}	yes	1.89e-18	753	754	754
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.85}	yes	5.84e-19	278	279	279
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.9}	yes	1.69e-18	407	408	408
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.95}	yes	1.07e-18	812	813	813
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.5}	yes	1.78e-18	375	376	376
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.7}	yes	1.66e-18	152	153	153
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.85}	yes	1.21e-18	258	259	259
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.9}	yes	8.42e-19	396	397	397
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.95}	yes	4.01e-19	818	819	819
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.5}	yes	1.33e-18	92	93	93
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.7}	yes	6.23e-19	124	125	125
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.85}	yes	4.13e-19	259	260	260
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.9}	yes	3.36e-19	402	403	403
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.95}	yes	1.74e-18	808	809	809
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.5}	yes	7.32e-19	63	64	64
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.7}	yes	2.00e-19	122	123	123
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.85}	yes	2.07e-19	255	256	256
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.9}	yes	6.48e-20	408	409	409

Таблица 30: Full Lab 3 experiment table: rows 865–896 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.95}	yes	3.63e-19	814	815	815
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.5}	no	5.17e-07	1500	1501	3001
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.7}	no	1.22e-11	1500	1501	3001
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.85}	yes	1.90e-18	1103	1104	2207
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.9}	yes	1.90e-18	546	547	1093
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.95}	yes	1.11e-18	769	770	1539
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.5}	yes	1.93e-18	1335	1336	2671
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.7}	yes	1.87e-18	769	770	1539
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.85}	yes	1.44e-18	250	251	501
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.9}	yes	4.95e-19	348	349	697
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.95}	yes	1.25e-18	627	628	1255
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.5}	yes	1.88e-18	386	387	773
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.7}	yes	1.59e-18	185	186	371
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.85}	yes	1.66e-18	198	199	397
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.9}	yes	7.65e-19	284	285	569
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.95}	yes	7.63e-19	450	451	901
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.5}	yes	1.63e-18	111	112	223
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.7}	yes	1.44e-18	83	84	167
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.85}	yes	4.22e-20	160	161	321
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.9}	yes	1.48e-18	212	213	425
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.95}	yes	9.18e-20	210	211	421
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.5}	yes	6.84e-20	43	44	87
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.7}	yes	1.16e-18	61	62	123
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.85}	yes	2.30e-19	88	89	177
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.9}	yes	3.88e-19	93	94	187
Himmelblau	[2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.95}	yes	4.47e-19	116	117	233
Himmelblau	[2.0, 2.0]	AdaGrad	{'alpha': 0.001}	no	21.74	1500	1501	1501
Himmelblau	[2.0, 2.0]	AdaGrad	{'alpha': 0.003}	no	15.02	1500	1501	1501
Himmelblau	[2.0, 2.0]	AdaGrad	{'alpha': 0.01}	no	3.423	1500	1501	1501
Himmelblau	[2.0, 2.0]	AdaGrad	{'alpha': 0.03}	no	6.01e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	AdaGrad	{'alpha': 0.1}	yes	9.29e-19	550	551	551
Himmelblau	[2.0, 2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.5}	no	17.92	1500	1501	1501

Таблица 31: Full Lab 3 experiment table: rows 897–928 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.7}	no	17.92	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.85}	no	17.9	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.9}	no	17.89	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.95}	no	17.84	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.5}	no	7.801	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.7}	no	7.797	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.85}	no	7.783	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.9}	no	7.767	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.95}	no	7.713	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.5}	no	1.85e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.7}	no	1.85e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.85}	no	1.85e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.9}	no	1.85e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.001, 'rho': 0.95}	no	1.85e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.5}	no	1.67e-04	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.7}	no	1.66e-04	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.85}	no	1.67e-04	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.9}	no	1.67e-04	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.95}	no	1.67e-04	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.5}	no	0.001856	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.7}	no	0.001856	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.85}	no	0.001844	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.9}	no	0.001844	1500	1501	1501
Himmelblau	[2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.95}	yes	3.31e-19	187	188	188
Himmelblau	[2.0, 2.0]	AdaDelta	{'rho': 0.5}	no	1.02e-04	1500	1501	1501
Himmelblau	[2.0, 2.0]	AdaDelta	{'rho': 0.7}	no	5.51e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	AdaDelta	{'rho': 0.85}	no	0.006154	1500	1501	1501
Himmelblau	[2.0, 2.0]	AdaDelta	{'rho': 0.9}	no	0.2069	1500	1501	1501
Himmelblau	[2.0, 2.0]	AdaDelta	{'rho': 0.95}	no	1.012	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	17.93	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	17.94	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	18.01	1500	1501	1501

Таблица 32: Full Lab 3 experiment table: rows 929–960 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	18.08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	18.22	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	17.93	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	17.94	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	18.01	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	18.08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	18.22	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	17.93	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	17.94	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	18.01	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	18.08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	18.22	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	17.92	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	17.93	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	18.01	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	18.07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	18.21	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	17.91	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	17.92	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	18	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	18.06	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	18.21	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	7.828	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	7.865	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	8.135	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	8.378	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	8.852	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	7.822	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	7.859	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	8.129	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	8.373	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	8.847	1500	1501	1501

Таблица 33: Full Lab 3 experiment table: rows 961–992 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	7.815	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	7.852	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	8.123	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	8.368	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	8.843	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	7.803	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	7.84	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	8.111	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	8.358	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	8.834	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	7.765	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	7.803	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	8.077	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	8.328	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	8.808	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	2.49e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	3.77e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	1.41e-09	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	0.002147	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	0.1398	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	1.65e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	6.35e-08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	2.13e-10	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	0.001904	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	0.1368	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	1.34e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	2.29e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	5.32e-12	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	0.001679	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	0.1338	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	4.01e-08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2'...	no	1.83e-08	1500	1501	1501

Таблица 34: Full Lab 3 experiment table: rows 993–1024 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	8.44e-13	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	0.001281	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	0.1279	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	1.11e-08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	2.64e-08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	3.44e-10	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	4.56e-04	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	0.1112	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	1.28e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	1.26e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	8.38e-19	818	819	819
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	9.56e-19	1129	1130	1130
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	9.56e-13	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	5.31e-06	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	9.24e-08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	4.59e-19	764	765	765
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	9.45e-19	1099	1100	1100
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	4.08e-13	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	1.34e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	2.35e-06	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	yes	4.80e-19	796	797	797
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	yes	8.48e-19	1059	1060	1060
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	1.58e-13	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	3.49e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	2.02e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	yes	3.58e-19	794	795	795
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	yes	6.62e-19	1010	1011	1011
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	1.56e-14	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	2.60e-08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	no	3.51e-09	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	yes	4.51e-20	1017	1018	1018
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2':...	yes	9.67e-19	1145	1146	1146

Таблица 35: Full Lab 3 experiment table: rows 1025–1056 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2': ...}	yes	9.67e-19	1443	1444	1444
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	5.36e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	yes	2.57e-20	237	238	238
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	yes	6.02e-19	371	372	372
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	yes	1.00e-18	468	469	469
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	yes	9.72e-19	581	582	582
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	1.22e-05	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	3.55e-06	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	yes	7.34e-19	366	367	367
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	yes	1.88e-19	408	409	409
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	yes	1.01e-18	536	537	537
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	no	7.99e-06	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	no	8.70e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	yes	6.87e-19	367	368	368
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	yes	8.76e-19	420	421	421
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2': ...}	yes	3.95e-19	454	455	455
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	no	7.06e-08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	no	6.80e-08	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	yes	1.18e-18	459	460	460
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	yes	4.15e-19	447	448	448
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...}	yes	5.77e-19	425	426	426
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	no	2.50e-06	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	no	5.73e-07	1500	1501	1501
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	yes	1.58e-18	827	828	828
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	yes	7.30e-20	838	839	839
Himmelblau	[2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2': ...}	yes	7.04e-19	865	866	866
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.5}	no	5.263	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.7}	no	4.605	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.85}	no	2.188	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.9}	no	0.2077	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0001, 'beta': 0.95}	no	0.006153	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.5}	no	3.266	1500	1501	1501

Таблица 36: Full Lab 3 experiment table: rows 1057–1088 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.7}	no	0.4612	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.85}	no	0.01969	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.9}	no	0.003019	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.0003, 'beta': 0.95}	no	3.96e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.5}	no	0.004159	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.7}	no	0.002348	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.85}	no	2.60e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.9}	no	5.05e-07	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.001, 'beta': 0.95}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.5}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.7}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.85}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.9}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.003, 'beta': 0.95}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.5}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.7}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.85}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.9}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Momentum	{'alpha': 0.01, 'beta': 0.95}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.5}	no	5.277	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.7}	no	4.668	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.85}	no	2.78	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.9}	no	0.4993	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.95}	no	0.03005	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.5}	no	3.307	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.7}	no	0.5783	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.85}	no	0.03497	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.9}	no	0.0059	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.95}	no	4.33e-05	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.5}	no	0.008522	1500	1501	3001
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.7}	–	–	–	–	–
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.85}	–	–	–	–	–

Таблица 37: Full Lab 3 experiment table: rows 1089–1120 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.9}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.001, 'beta': 0.95}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.5}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.7}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.85}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.9}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.003, 'beta': 0.95}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.5}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.7}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.85}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.9}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	Nesterov	{'alpha': 0.01, 'beta': 0.95}	—	—	—	—	—
Rosenbrock	[-2.0, 2.0]	AdaGrad	{'alpha': 0.001}	no	277.6	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	AdaGrad	{'alpha': 0.003}	no	116.3	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	AdaGrad	{'alpha': 0.01}	no	7.396	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	AdaGrad	{'alpha': 0.03}	no	6.544	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	AdaGrad	{'alpha': 0.1}	no	6.181	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.5}	no	170	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.7}	no	170	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.85}	no	169.7	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.9}	no	169.5	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0001, 'rho': 0.95}	no	168.7	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.5}	no	6.538	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.7}	no	6.541	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.85}	no	6.552	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.9}	no	6.562	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.0003, 'rho': 0.95}	no	6.646	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.5}	no	4.754	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.7}	no	4.768	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.85}	no	4.801	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.9}	no	4.837	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSPProp	{'alpha': 0.001, 'rho': 0.95}	no	4.955	1500	1501	1501

Таблица 38: Full Lab 3 experiment table: rows 1121–1152 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.5}	no	0.09104	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.7}	no	0.0879	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.85}	no	0.08793	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.9}	no	0.09337	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.003, 'rho': 0.95}	no	0.1241	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.5}	no	0.0547	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.7}	no	0.03963	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.85}	no	0.03482	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.9}	no	0.03379	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	RMSProp	{'alpha': 0.01, 'rho': 0.95}	no	0.03467	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	AdaDelta	{'rho': 0.5}	no	4.076	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	AdaDelta	{'rho': 0.7}	no	5.082	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	AdaDelta	{'rho': 0.85}	no	5.892	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	AdaDelta	{'rho': 0.9}	no	6.194	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	AdaDelta	{'rho': 0.95}	no	6.494	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	170.5	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	171	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	175.2	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	179.1	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	187.3	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	170.4	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	170.9	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	175.1	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	179	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	187.2	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	170.3	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	170.8	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	175	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	179	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	187.2	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	170	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	170.6	1500	1501	1501

Таблица 39: Full Lab 3 experiment table: rows 1153–1184 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	174.9	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	178.8	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	187	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	169.4	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	170.1	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	174.3	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	178.3	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	186.5	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	6.559	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	6.714	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	11.16	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	17.58	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	32.01	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	6.555	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	6.674	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	11.04	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	17.44	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	31.87	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	6.552	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	6.64	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	10.92	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	17.3	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	31.73	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	6.549	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	6.588	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	10.68	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	17.03	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	31.45	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	6.553	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	6.565	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	9.981	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	16.22	1500	1501	1501

Таблица 40: Full Lab 3 experiment table: rows 1185–1216 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	30.63	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2':...	no	4.815	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2':...	no	4.961	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2':...	no	6.203	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2':...	no	6.541	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2':...	no	6.584	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	no	4.806	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	no	4.953	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	no	6.2	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	no	6.54	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2':...	no	6.584	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	4.798	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	4.945	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	6.196	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	6.539	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	6.584	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	4.794	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	4.931	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	6.188	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	6.538	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	6.583	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	4.906	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	4.93	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	6.165	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	6.535	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2'...	no	6.581	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	0.006477	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	0.01206	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	3.89	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	6.211	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	6.509	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	0.003343	1500	1501	1501

Таблица 41: Full Lab 3 experiment table: rows 1217–1248 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	0.01037	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	3.875	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	6.207	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	6.507	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2'...	no	0.002472	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2'...	no	0.01044	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2'...	no	3.861	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2'...	no	6.203	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2'...	no	6.506	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	0.002983	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	0.01997	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	3.833	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	6.195	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	6.503	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	no	0.2179	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	no	0.09234	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	no	3.768	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	no	6.173	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	no	6.495	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	1.60e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	1.44e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	0.003656	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	3.927	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...}	no	6.018	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	6.06e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	3.06e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	0.001049	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	3.876	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...}	no	6.005	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	no	1.41e-04	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	no	3.31e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	no	6.78e-04	1500	1501	1501

Таблица 42: Full Lab 3 experiment table: rows 1249–1280 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	no	3.834	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	no	5.994	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	1.59e-04	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	1.43e-04	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	4.07e-04	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	3.772	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	5.976	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	4.79e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	3.55e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	7.80e-05	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	3.727	1500	1501	1501
Rosenbrock	[-2.0, 2.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	5.941	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0001, 'beta': 0.5}	no	3.69e-16	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0001, 'beta': 0.7}	yes	7.61e-19	996	997	997
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0001, 'beta': 0.85}	yes	7.01e-19	304	305	305
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0001, 'beta': 0.9}	yes	5.10e-19	391	392	392
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0001, 'beta': 0.95}	yes	3.92e-19	799	800	800
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0003, 'beta': 0.5}	yes	7.60e-19	555	556	556
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0003, 'beta': 0.7}	yes	6.51e-19	277	278	278
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0003, 'beta': 0.85}	yes	2.17e-19	257	258	258
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0003, 'beta': 0.9}	yes	3.29e-19	394	395	395
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.0003, 'beta': 0.95}	yes	2.27e-19	760	761	761
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.001, 'beta': 0.5}	yes	5.72e-19	136	137	137
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.001, 'beta': 0.7}	yes	6.62e-19	116	117	117
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.001, 'beta': 0.85}	yes	6.59e-19	256	257	257
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.001, 'beta': 0.9}	yes	2.04e-20	387	388	388
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.001, 'beta': 0.95}	yes	1.65e-19	801	802	802
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.003, 'beta': 0.5}	yes	1.41e-20	63	64	64
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.003, 'beta': 0.7}	yes	2.32e-19	116	117	117
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.003, 'beta': 0.85}	yes	2.97e-19	252	253	253
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.003, 'beta': 0.9}	yes	3.89e-19	385	386	386
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.003, 'beta': 0.95}	yes	4.93e-19	814	815	815

Таблица 43: Full Lab 3 experiment table: rows 1281–1312 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.01, 'beta': 0.5}	yes	2.28e-19	60	61	61
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.01, 'beta': 0.7}	yes	4.19e-19	118	119	119
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.01, 'beta': 0.85}	yes	2.85e-19	250	251	251
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.01, 'beta': 0.9}	yes	4.80e-19	390	391	391
Himmelblau	[-2.0, 3.0]	Momentum	{'alpha': 0.01, 'beta': 0.95}	yes	2.30e-19	747	748	748
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.5}	no	4.75e-16	1500	1501	3001
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.7}	yes	7.46e-19	1013	1014	2027
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.85}	yes	7.13e-19	339	340	679
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.9}	yes	6.36e-19	400	401	801
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0001, 'beta': 0.95}	yes	4.42e-19	687	688	1375
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.5}	yes	7.53e-19	567	568	1135
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.7}	yes	7.45e-19	298	299	597
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.85}	yes	7.06e-19	241	242	483
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.9}	yes	2.13e-19	309	310	619
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.0003, 'beta': 0.95}	yes	1.09e-19	540	541	1081
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.001, 'beta': 0.5}	yes	6.64e-19	152	153	305
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.001, 'beta': 0.7}	yes	2.20e-20	100	101	201
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.001, 'beta': 0.85}	yes	2.34e-19	176	177	353
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.001, 'beta': 0.9}	yes	1.65e-19	220	221	441
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.001, 'beta': 0.95}	yes	1.74e-20	314	315	629
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.003, 'beta': 0.5}	yes	4.88e-20	50	51	101
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.003, 'beta': 0.7}	yes	7.61e-19	73	74	147
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.003, 'beta': 0.85}	yes	3.21e-19	109	110	219
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.003, 'beta': 0.9}	yes	5.43e-20	115	116	231
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.003, 'beta': 0.95}	yes	4.30e-19	156	157	313
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.01, 'beta': 0.5}	yes	7.30e-21	24	25	49
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.01, 'beta': 0.7}	yes	4.30e-19	32	33	65
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.01, 'beta': 0.85}	yes	2.17e-20	36	37	73
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.01, 'beta': 0.9}	yes	5.29e-19	36	37	73
Himmelblau	[-2.0, 3.0]	Nesterov	{'alpha': 0.01, 'beta': 0.95}	yes	9.25e-21	39	40	79
Himmelblau	[-2.0, 3.0]	AdaGrad	{'alpha': 0.001}	no	13.26	1500	1501	1501
Himmelblau	[-2.0, 3.0]	AdaGrad	{'alpha': 0.003}	no	8.93	1500	1501	1501

Таблица 44: Full Lab 3 experiment table: rows 1313–1344 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[-2.0, 3.0]	AdaGrad	{'alpha': 0.01}	no	0.9643	1500	1501	1501
Himmelblau	[-2.0, 3.0]	AdaGrad	{'alpha': 0.03}	no	1.31e-08	1500	1501	1501
Himmelblau	[-2.0, 3.0]	AdaGrad	{'alpha': 0.1}	yes	6.85e-19	331	332	332
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.5}	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.7}	no	10.94	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.85}	no	10.93	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.9}	no	10.92	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0001, 'rho': 0.95}	no	10.89	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.5}	no	3.605	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.7}	no	3.602	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.85}	no	3.592	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.9}	no	3.58	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.0003, 'rho': 0.95}	no	3.542	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.001, 'rho': 0.5}	no	1.85e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.001, 'rho': 0.7}	no	1.85e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.001, 'rho': 0.85}	no	1.85e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.001, 'rho': 0.9}	no	1.85e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.001, 'rho': 0.95}	no	1.85e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.003, 'rho': 0.5}	no	1.67e-04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.003, 'rho': 0.7}	no	1.66e-04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.003, 'rho': 0.85}	no	1.66e-04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.003, 'rho': 0.9}	no	1.66e-04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.003, 'rho': 0.95}	no	1.66e-04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.01, 'rho': 0.5}	no	0.001849	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.01, 'rho': 0.7}	no	0.001849	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.01, 'rho': 0.85}	no	0.001849	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.01, 'rho': 0.9}	no	0.001849	1500	1501	1501
Himmelblau	[-2.0, 3.0]	RMSProp	{'alpha': 0.01, 'rho': 0.95}	no	0.001851	1500	1501	1501
Himmelblau	[-2.0, 3.0]	AdaDelta	{'rho': 0.5}	no	1.49e-04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	AdaDelta	{'rho': 0.7}	no	1.46e-04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	AdaDelta	{'rho': 0.85}	no	1.14e-04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	AdaDelta	{'rho': 0.9}	no	3.22e-04	1500	1501	1501

Таблица 45: Full Lab 3 experiment table: rows 1345–1376 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[-2.0, 3.0]	AdaDelta	{'rho': 0.95}	no	0.09939	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	10.97	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	10.99	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.7, 'beta2'...	no	11.04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	10.97	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	10.99	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.8, 'beta2'...	no	11.04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	10.97	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	10.99	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.85, 'beta2'...	no	11.04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	10.97	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	10.99	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.9, 'beta2'...	no	11.04	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	10.95	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	10.97	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	10.99	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0001, 'beta1': 0.95, 'beta2'...	no	11.03	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	3.625	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	3.654	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	3.848	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	4.021	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.7, 'beta2'...	no	4.364	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	3.62	1500	1501	1501

Таблица 46: Full Lab 3 experiment table: rows 1377–1408 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	3.649	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	3.844	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	4.017	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.8, 'beta2'...	no	4.36	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	3.615	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	3.644	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	3.839	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	4.013	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.85, 'beta2'...	no	4.357	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	3.606	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	3.634	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	3.83	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	4.005	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.9, 'beta2'...	no	4.35	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	3.577	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	3.606	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	3.804	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	3.981	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.0003, 'beta1': 0.95, 'beta2'...	no	4.329	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	5.43e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	5.49e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	yes	7.22e-19	1390	1391	1391
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	3.36e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.7, 'beta2'...	no	0.004593	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	2.17e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	1.03e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	yes	4.12e-20	1344	1345	1345
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	2.15e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.8, 'beta2'...	no	0.004355	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	1.67e-08	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	no	5.13e-08	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2'...	yes	2.08e-21	1339	1340	1340

Таблица 47: Full Lab 3 experiment table: rows 1409–1440 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	no	1.29e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.85, 'beta2':...	no	0.004124	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	8.81e-08	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	5.68e-08	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	yes	1.03e-19	1372	1373	1373
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	3.67e-08	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.9, 'beta2':...	no	0.003684	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	6.22e-09	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	1.29e-08	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	2.25e-17	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	1.26e-09	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.001, 'beta1': 0.95, 'beta2':...	no	0.002538	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	2.51e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	no	2.53e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	7.21e-19	703	704	704
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	6.75e-19	946	947	947
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.7, 'beta2':...	yes	7.63e-19	1396	1397	1397
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	2.35e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	no	2.47e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	7.17e-19	681	682	682
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	7.13e-19	914	915	915
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.8, 'beta2':...	yes	7.37e-19	1365	1366	1366
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	4.41e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	no	6.66e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	yes	2.59e-20	643	644	644
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	yes	6.91e-19	868	869	869
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.85, 'beta2':...	yes	7.26e-19	1331	1332	1332
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	6.22e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	no	1.62e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	yes	7.66e-20	715	716	716
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	yes	6.25e-20	814	815	815
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.9, 'beta2':...	yes	7.15e-19	1250	1251	1251

Таблица 48: Full Lab 3 experiment table: rows 1441–1470 of 1470

Function	x_0	Optimizer	Params	conv	f	iter	N_f	N_g
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	no	3.86e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	no	2.63e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	yes	1.42e-20	998	999	999
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	yes	3.49e-20	1014	1015	1015
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.003, 'beta1': 0.95, 'beta2'...	yes	3.15e-19	1069	1070	1070
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	no	3.76e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	no	1.22e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	2.32e-20	284	285	285
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	6.50e-19	352	353	353
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.7, 'beta2': ...	yes	7.68e-19	415	416	416
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	no	2.34e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	no	1.62e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	2.78e-19	281	282	282
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	3.13e-19	309	310	310
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.8, 'beta2': ...	yes	6.84e-19	348	349	349
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	no	1.25e-05	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	no	3.30e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	1.12e-20	283	284	284
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	4.91e-19	293	294	294
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.85, 'beta2':...	yes	6.91e-19	346	347	347
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	6.37e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	no	8.78e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	yes	1.87e-19	403	404	404
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	yes	1.26e-19	407	408	408
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.9, 'beta2': ...	yes	6.09e-19	422	423	423
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	1.09e-06	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	no	1.73e-07	1500	1501	1501
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	yes	5.06e-19	841	842	842
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	yes	5.62e-19	790	791	791
Himmelblau	[-2.0, 3.0]	Adam	{'alpha': 0.01, 'beta1': 0.95, 'beta2':...	yes	3.41e-19	821	822	822

7 Общий вывод

В работе были реализованы шесть модификаций градиентного спуска. Эксперименты показали, что инерция и адаптивное масштабирование шага действительно могут ускорить сходимость по сравнению с базовым градиентным спуском, но за это приходится внимательнее подбирать гиперпараметры.

Momentum и Nesterov хорошо проявили себя на квадратичных функциях. Для обеих квадратичных моделей удачными оказались параметры порядка $\alpha = 10^{-2}$ и $\beta = 0.85$. Метод Нестерова иногда требовал меньше итераций на плохо обусловленной функции, однако платил за это дополнительными вычислениями градиента.

RMSProp показал устойчивую работу на квадратичных функциях и подтвердил полезность экспоненциального сглаживания квадратов градиента. AdaGrad и AdaDelta в выбранной сетке часто двигались слишком осторожно и не достигали точности до лимита итераций.

Adam показал себя как удобный компромисс между инерцией и адаптивным шагом. При этом на детерминированных тестовых задачах он не всегда был самым быстрым: на простых квадратичных функциях Momentum и RMSProp часто требовали меньше итераций.

На сложных функциях результат сильнее зависел от стартовой точки и геометрии поверхности. Функция Розенброка осталась наиболее трудной из-за узкой изогнутой долины, а функция Химмельблау хорошо решалась из стартовых точек, попадающих в области притяжения глобальных минимумов.